Enterprise Network Design and Implementation for LankaTrade Holdings (Pvt) Ltd using Cisco Packet Tracer

Project Scenario

LankaTrade Holdings (Pvt) Ltd is a rapidly growing Sri Lankan trading and distribution company headquartered in Colombo. The company specializes in import, export, logistics, and business support services across Sri Lanka. Due to continued expansion and an increasing workforce, the company has acquired a new office building to serve as its main operational headquarters.

The new facility currently has no existing network infrastructure. Therefore, a secure, scalable, and highly available enterprise network must be designed and implemented to support the organization's current business operations and future growth.

The headquarters building consists of three floors with the following departments:

First Floor
• Sales and Marketing Department (120 users)
• Human Resources and Logistics Department (120 users)

Second Floor
• Finance and Accounts Department (120 users)
• Administration and Public Relations Department (120 users)

Third Floor
• Information and Communication Technology (ICT) Department (120 users)
• Server Room (12 devices)

As a Network Engineer working with the Infrastructure Team of LankaTrade Holdings (Pvt) Ltd, you have been tasked with designing and implementing a reliable enterprise network solution that follows industry best practices for security, scalability, redundancy, and performance.

Project Requirements

• Use Cisco Packet Tracer to design and implement the network solution.
• Implement a hierarchical network design model with redundancy at every layer.
• Deploy two routers and two multilayer switches to provide network redundancy and high availability.
• Connect the enterprise network to two Internet Service Providers (ISPs) for Internet redundancy and failover capabilities.
• Provide wireless network access for users in every department.
• Place each department in a separate VLAN and separate IP subnet.
• Using the base network 172.16.1.0, perform subnetting to allocate appropriate IP address ranges for each department.
• Configure the enterprise network to use the following public IPv4 ranges: 195.136.17.0/30, 195.136.17.4/30, 195.136.17.8/30, and 195.136.17.12/30.
• Configure hostnames, console passwords, enable passwords, login banners, and disable IP domain lookup.
• Ensure all departments can communicate through inter-VLAN routing configured on multilayer switches.
• Configure multilayer switches to perform both switching and routing functions.
• Deploy dedicated DHCP servers in the Server Room for automatic IP allocation.

• Configure static IP addressing for all server room devices.
• Implement OSPF as the dynamic routing protocol on routers and multilayer switches.
• Configure SSH on all routers and multilayer switches for secure remote administration.
• Implement switch port security within the Finance and Accounts Department using sticky MAC address learning, one device per port, and shutdown violation mode.
• Configure PAT (Port Address Translation) using outbound router interface IP addresses.
• Implement the necessary Access Control Lists (ACLs) to support PAT operations.
• Test and verify end-to-end network connectivity and functionality.

Technologies Implemented

• Creating a network topology using Cisco Packet Tracer
• Hierarchical Network Design
• Connecting networking devices using appropriate cabling
• Configuring basic device settings
• Creating VLANs and assigning switch ports
• Subnetting and IP Addressing
• Inter-VLAN Routing using Multilayer Switches (SVI)
• Dedicated DHCP Server Configuration
• SSH Secure Remote Access
• OSPF Dynamic Routing Protocol
• NAT Overload (PAT)
• Standard and Extended ACL Configuration
• Switch Port Security
• Wireless Network (WLAN) Implementation
• Host Device Configuration
• ISP Router Configuration
• Network Testing and Verification